

USING MIDI WITH CIRCUITPYTHON

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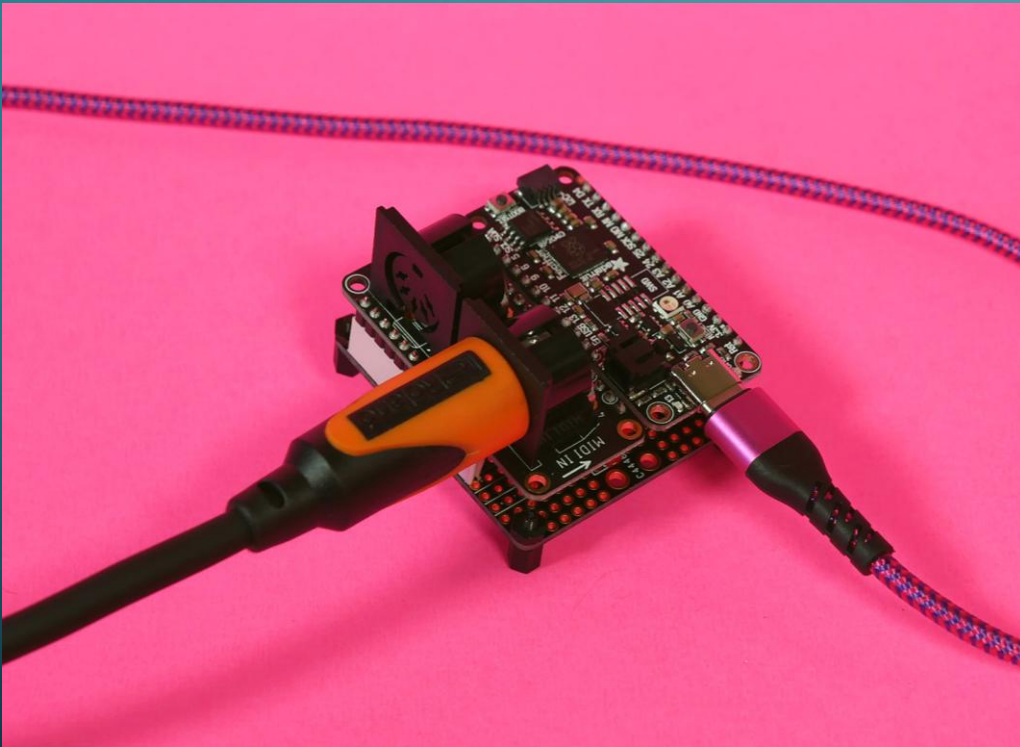


WHAT IS MIDI?

- Acronym: Music Instrument Digital Interface
- 7-bit communication protocol
 - Dates back to the 1980's
- Dependable, predictable, reliable

A white rectangular box containing the word "MIDI" in a bold, black, stylized font. The letters are thick and blocky, with the 'I' being a simple vertical bar. The 'M' and 'D' have unique internal cutouts.

HOW MIDI IS TRANSMITTED



- Bidirectional
 - MIDI in: **receiving** MIDI messages as an **input**
 - MIDI out: **sending** MIDI messages as an **output**
- MIDI Channels
 - Messages are sent via channels, 1-16
 - Allow devices to know what and who to listen to

CONNECTION TYPES



DIN-5

- The original MIDI connector
- Different circuit for MIDI in and MIDI out



TRS-A/TRS-B

- MIDI via an 1/8" audio jack
- Small form factor, but not always standardized
- Different circuit for MIDI in and MIDI out



USB

- Plug and play
- Bidirectional
- You'll be using this today

BLE MIDI

- Wireless MIDI transmission
- Support is available in CircuitPython
 - Can be used on the Circuit Playground Bluefruit



MIDI MESSAGES

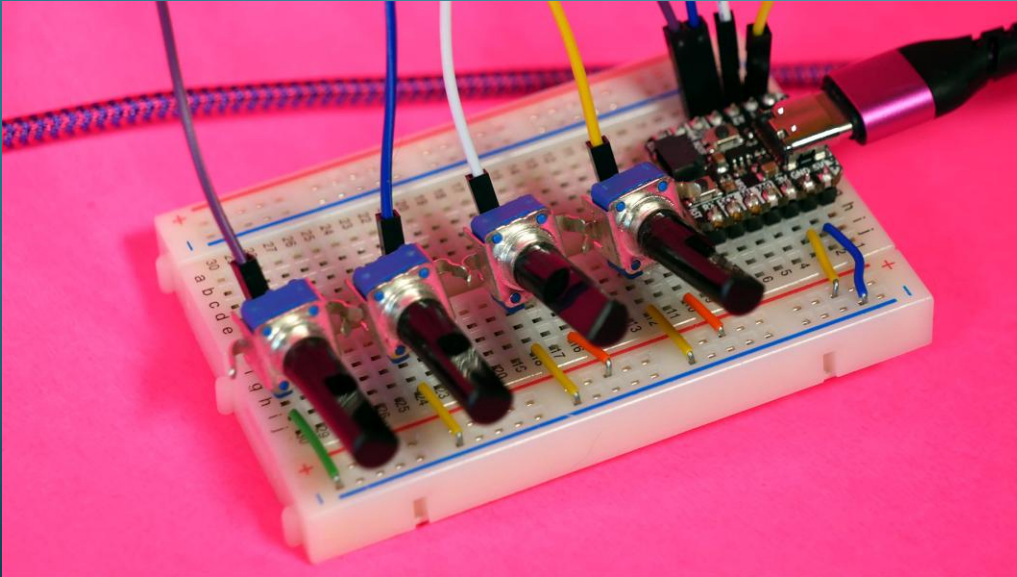


NOTEON AND NOTEOFF

- How notes are played
 - i.e. when you play a piano keyboard
- Notes in MIDI are numbered 1-127
 - Ex: C4/Middle C is 60
 - MIDI NOTE NUMBERS AND CENTER FREQUENCIES
- Velocity: strength, force, loudness



CONTROL CHANGE (CC)

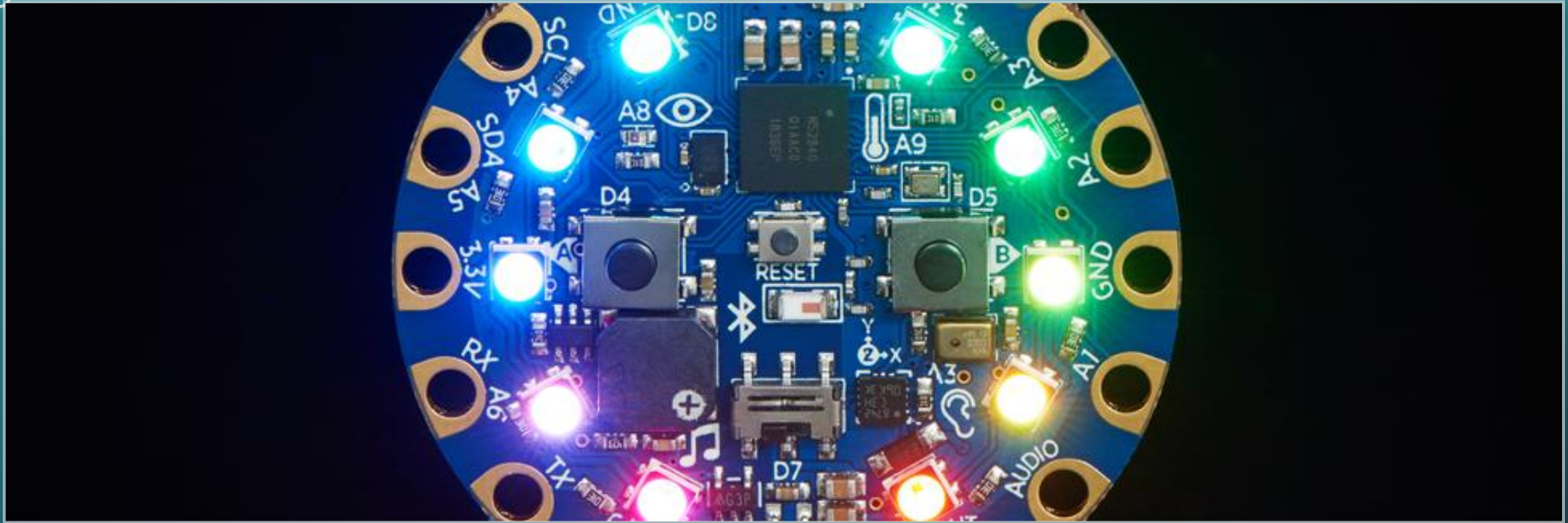


- Utility messages, functionality varies
 - Usually sent by a knob or a slider
- Common examples: sustain, modulation, breath control
 - [CC Message table](#)
- CC message contains the message number and the value (0-127)

PITCHBEND

- Special MIDI message, but very common
- Unique range 0-16383, with 8192 as the midrange/no bend

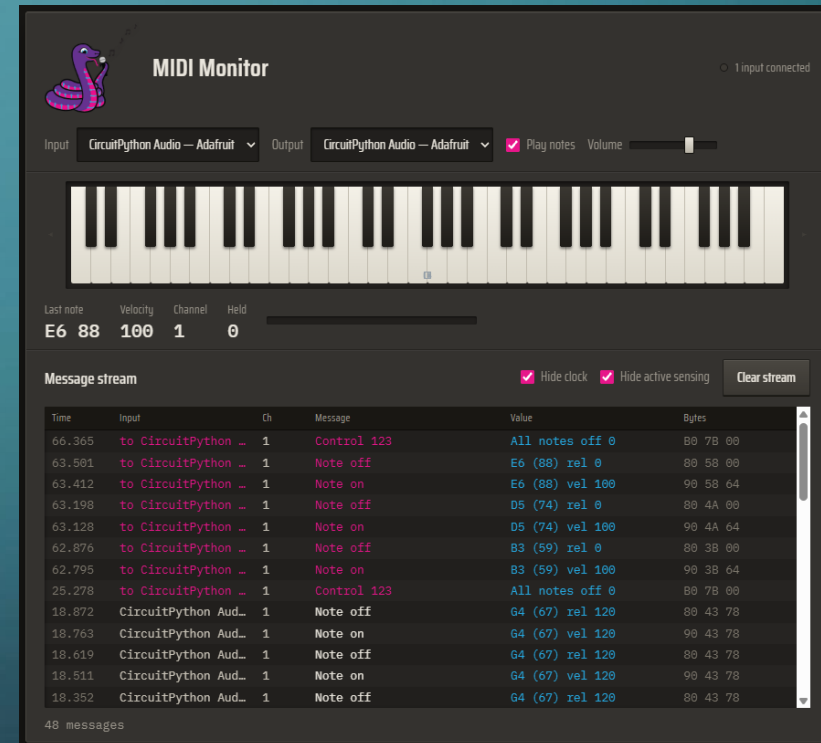




LET'S BUILD!

MIDI WITH CIRCUITPYTHON LAB

- Using the Circuit Playground Bluefruit
- Interface with the [MIDI Monitor](#) for debugging and testing
 - Runs in your browser (Chrome, Firefox, Edge) and OS agnostic



The screenshot shows the MIDI Monitor web interface. At the top, there's a logo of a purple snake and the title "MIDI Monitor". Below it, there are dropdown menus for "Input" and "Output", both set to "CircuitPython Audio — Adafruit". There are checkboxes for "Play notes" (checked) and "Volume" (slider). A piano roll is visible in the center. Below the piano roll, there's a section for "Last note" with fields for Note (E6), Pitch (88), Velocity (100), Channel (1), and Held (0). At the bottom, there's a "Message stream" section with checkboxes for "Hide clock" (checked) and "Hide active sensing" (checked), and a "Clear stream" button. The message stream table shows the following data:

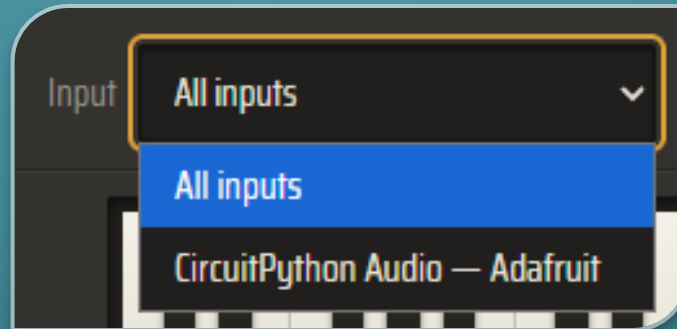
Time	Input	Ch	Message	Value	Bytes
66.365	to CircuitPython _	1	Control 123	All notes off 0	80 78 00
63.501	to CircuitPython _	1	Note off	E6 (88) rel 0	80 58 00
63.412	to CircuitPython _	1	Note on	E6 (88) vel 100	90 58 64
63.198	to CircuitPython _	1	Note off	D5 (74) rel 0	80 4A 00
63.128	to CircuitPython _	1	Note on	D5 (74) vel 100	90 4A 64
62.876	to CircuitPython _	1	Note off	B3 (59) rel 0	80 38 00
62.795	to CircuitPython _	1	Note on	B3 (59) vel 100	90 38 64
25.278	to CircuitPython _	1	Control 123	All notes off 0	80 78 00
18.872	CircuitPython Aud_	1	Note off	G4 (67) rel 120	80 43 78
18.763	CircuitPython Aud_	1	Note on	G4 (67) vel 120	90 43 78
18.619	CircuitPython Aud_	1	Note off	G4 (67) rel 120	80 43 78
18.511	CircuitPython Aud_	1	Note on	G4 (67) vel 120	90 43 78
18.352	CircuitPython Aud_	1	Note off	G4 (67) rel 120	80 43 78

48 messages

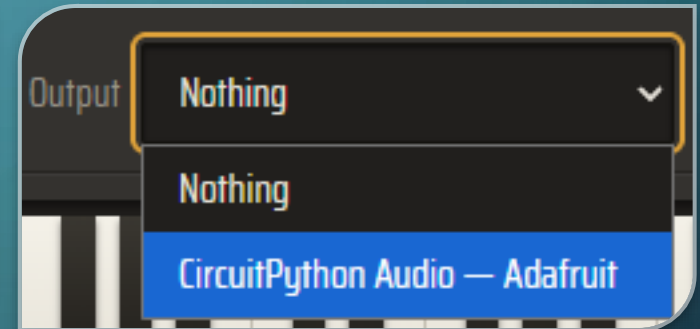
HOW TO CONNECT



CLICK CONNECT A MIDI
DEVICE



SELECT INPUT



SELECT OUTPUT

1. NEOPIXEL MIDI LISTENER

- Create a MIDI input device
- The code should listen for NoteOn and NoteOff messages
- The message type and value should print to the Serial Console
- When a message is received, the NeoPixels should light up
 - Each message should have a color associated with it
- For testing, connect to the MIDI Monitor as an output and click on the keyboard notes

2. CAPACITIVE TOUCH MIDI KEYBOARD

- Use the seven capacitive touch pads to send NoteOn and NoteOff messages
 - NoteOn should send when touched, NoteOff should send when released
- Start on C4
- For testing, connect to the MIDI Monitor as an input and watch the message stream
 - You will also be able to hear the notes if Play Notes is checked off
- Bonus: light up the NeoPixels a different color for each NoteOn message, turn off for NoteOff

2B. CHANGE OCTAVES

- Build on the cap touch keyboard code:
- Use the D4 and D5 buttons to go up or down an octave for the notes that are sent
- For testing, connect to the *MIDI Monitor* as an input and watch the message stream

3. SEND CC MESSAGES WITH THE LIGHT SENSOR

- Send Modulation CC values to the MIDI Monitor
- Use `simpleio map_range` to map the light sensor values to the CC range (0-127)
 - Hint: it may be helpful to run light sensor code first to get a better idea of the value ranges before adding MIDI
- For testing, connect to the MIDI Monitor as an input and watch the message stream

4. SEND PITCH BEND WITH THE ACCELEROMETER

- Send Pitch Bend values to the MIDI Monitor
- Use the Y axis value to control the Pitch Bend amount
- You can click on a note on the keyboard in the MIDI Monitor and hear the pitch bend up or down
- Remember: PitchBend values are not the standard 0-127 range, its 0-16383 with 8192 being center/no bend

5. BLE MIDI

- Choose your favorite demo so far and change the MIDI init at the top of the code from USB to BLE
- Won't try it in class because that many BLE devices might cause connection issues, harder to test (need music software)